Commitment Schemes and Zero-Knowledge Proofs

CS 171: Cryptography

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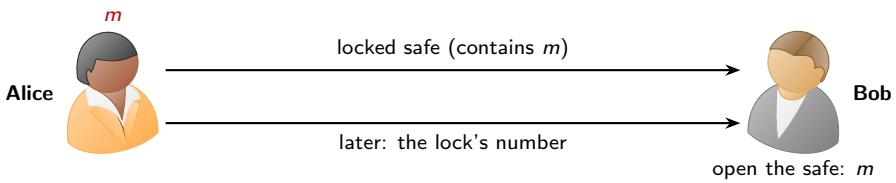
Overview

1. Commitment Schemes
2. Constructions
3. Statements and Witnesses
4. Zero-Knowledge Proof Systems
5. Three-Coloring in Zero-Knowledge
6. Zero-Knowledge: The Simulator
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Commitment

Goal

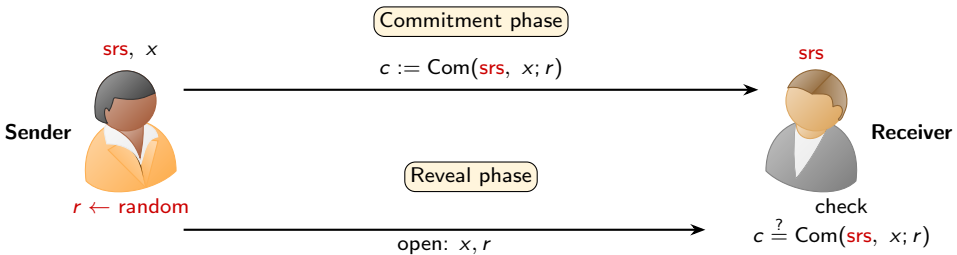
Bind to a secret value that **cannot later be explained with an alternate value** — like locking a message in a safe and handing it over.



Binding: cannot open to $m' \neq m$

Hiding: m stays secret until then

Commitment: Two Phases



- Often both parties share a **reference string** $\mathrm{srs} \leftarrow \mathrm{Setup}(1^n)$, e.g. a group and generators.
- In general, each phase can involve several back-and-forth messages. **Correctness**: an honest opening is always accepted.

Security: Binding and Hiding

Game $\mathrm{Bind}_{\mathcal{A},\Pi}(n)$

1. $\mathsf{srs} \leftarrow \mathrm{Setup}(1^n)$. $\mathcal{A}$ is given 1^n and srs.
2. $\mathcal{A}$ outputs (x, r) and (x', r').
3. Output 1 if $x \neq x'$ and $\mathrm{Com}(\mathsf{srs}, x; r) = \mathrm{Com}(\mathsf{srs}, x'; r')$.

Game $\mathrm{Hide}_{\mathcal{A},\Pi}(n)$

1. $\mathsf{srs} \leftarrow \mathrm{Setup}(1^n)$. $\mathcal{A}$ is given 1^n and srs; it outputs x_0, x_1.
2. $b \leftarrow \{0,1\}$, $c := \mathrm{Com}(\mathsf{srs}, x_b; r)$ for a fresh r; $\mathcal{A}$ gets c, outputs b'.
3. Output 1 if $b' = b$.

Definition 1 (Secure commitment)

$\Pi = (\mathrm{Setup}, \mathrm{Com})$ is **binding** if for all PPT $\mathcal{A}$: $\Pr[\mathrm{Bind}_{\mathcal{A},\Pi}(n) = 1] \leq \mathrm{negl}(n)$, and **hiding** if for all PPT $\mathcal{A}$: $\Pr[\mathrm{Hide}_{\mathcal{A},\Pi}(n) = 1] \leq \frac{1}{2} + \mathrm{negl}(n)$.

- **Binding**: no cheating sender can open one commitment two ways. **Hiding**: c leaks nothing about x.

Perfect or Computational?

Definition 2 (Hiding)

At the end of the **commitment phase**, no adversarial receiver learns any information about the committed value x.

- Against unbounded receivers: **perfectly hiding** ($\Pr[\text{Hide} = 1] = \frac{1}{2}$). Against PPT receivers: **computationally hiding**.

Definition 3 (Binding)

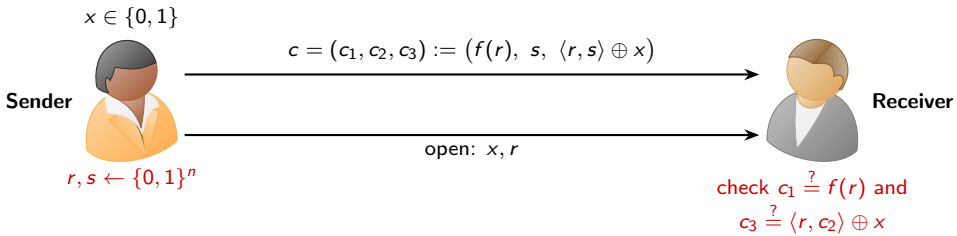
In the **reveal phase**, there is a *unique* value an adversarial sender can reveal as the value committed previously.

- Against unbounded senders: **perfectly binding**. Against PPT senders: **computationally binding**.

- A scheme cannot be **both** perfectly hiding and perfectly binding: if every x' has an opening of c, an unbounded sender can find one.

Commitment from Hardness Concentration

Let $f : \{0,1\}^n \to \{0,1\}^n$ be a one-way **permutation**; commit to one bit $x \in \{0,1\}$. (No srs.)



- **Binding** (perfect): f is a permutation, so c_1 determines r, and then $x = c_3 \oplus \langle r, c_2\rangle$ is determined.
- **Hiding** (computational): by Goldreich–Levin, $\langle r, s\rangle$ looks like a uniform bit given $f(r) \,\|\, s$ (lecture 12), so c_3 is a one-time pad on x.

Can We Use Any Encryption Scheme to Commit?

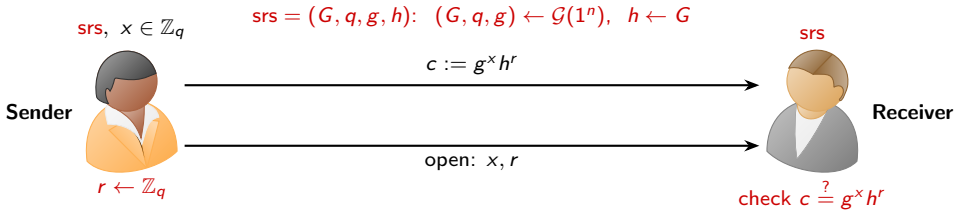
Given $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$: let $\mathsf{Com}(x; r)$ use r to run Gen, obtaining k, and output $\mathsf{Enc}_k(x)$; open by revealing r (and x).

- **No!**
- This commitment is hiding — but it need not be binding.
- Shouldn't binding come from the correctness of decryption? Correctness only holds for honestly chosen keys: a cheating sender may choose its coins dishonestly, e.g. find two keys $k \neq k'$ with $\mathsf{Enc}_k(x) = \mathsf{Enc}_{k'}(x')$.

Example

The one-time pad: $c = k \oplus x$ can be opened to *any* x' with the key $k' := c \oplus x'$.

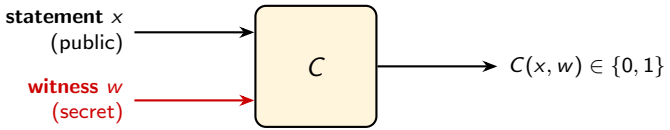
Pedersen Commitment



- **Binding** (computational): openings $(x, r) \neq (x', r')$ with $g^x h^r = c = g^{x'} h^{r'}$ give $\log_g h = (x - x')/(r' - r) \bmod q$ — the DLog hash of lecture 13 is exactly $(x, r) \mapsto g^x h^r$.
- **Hiding** (perfect): h^r is uniform in G, so c is uniform whatever x is. Indeed, for every $c = g^x h^r$ and every x', the value $r' = r + (x - x')/\log_g h$ gives $c = g^{x'} h^{r'}$.

Statements and Witnesses

A **circuit** C: a public, efficient program with two inputs and a one-bit output.



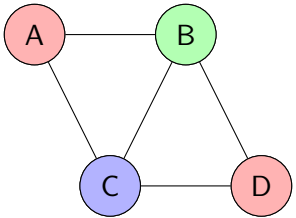
- The statement x is **true** if there *exists* a witness: $\exists w : \ C(x, w) = 1$.
- With w in hand, checking is easy: run C. Without w, finding one may be hard.

Examples

- **Discrete log**: $x = (G, q, g, h)$, $w = a \in \mathbb{Z}_q$; $C(x, a) = 1$ iff $g^a = h$. ("I know the secret key for h.")
- **Hash preimage**: $x = y$, $w = m$; $C(y, m) = 1$ iff $H(m) = y$.
- **Graph three-coloring**: next slide.

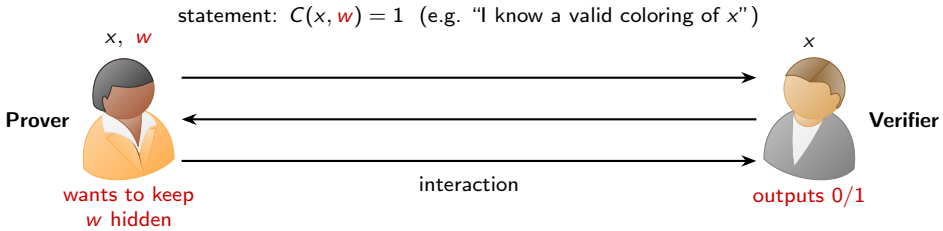
Graph Three-Coloring

- A graph $G = (V, E)$.
- Task: show a coloring $c : V \to \{R, G, B\}$ such that $c(u) \neq c(v)$ for all $(u, v) \in E$.



- Not every graph is three-colorable, and deciding whether a graph is three-colorable is believed to be computationally hard (it is NP-complete).
- **As a statement**: $x = G$, $w = c$, and $C(G, c) = 1$ iff $c(u) \neq c(v)$ for every $(u, v) \in E$ — just $|E|$ comparisons.

Zero-Knowledge Proof System



- The verifier should be convinced that $\exists w$ with $C(x, w) = 1$ ...
- ... and learn **nothing else** — in particular, nothing about w.

Zero-Knowledge Proofs: Definition

- **Syntax**: two PPT algorithms, the prover $P(1^n, x, w)$ and the verifier $V(1^n, x)$, which interact; V outputs 0 or 1.
- **Completeness**: an honest prover convinces an honest verifier: for all x, w with $C(x, w) = 1$,

$$\Pr\left[V \text{ outputs } 1 \text{ in } P(1^n, x, w) \leftrightarrow V(1^n, x)\right] = 1 - \mathrm{negl}(n).$$

- **Soundness**: a cheating prover cannot make the verifier accept a **false** statement: for all P^* and all x such that $C(x, w) = 0$ for every w,

$$\Pr\left[V \text{ outputs } 1 \text{ in } P^*(1^n, x) \leftrightarrow V(1^n, x)\right] = \mathrm{negl}(n).$$

- **Zero-knowledge**: the interaction leaks nothing about w ... next slide.

Zero-Knowledge: The Simulator

Idea: whatever a (possibly malicious) verifier V^* sees, it could have produced **without** the prover — by running a simulator $\mathcal{S}$ that does not know w.

Game $\mathrm{ZK}_{\mathcal{D},V^*,\mathcal{S}}(n,x,w)$

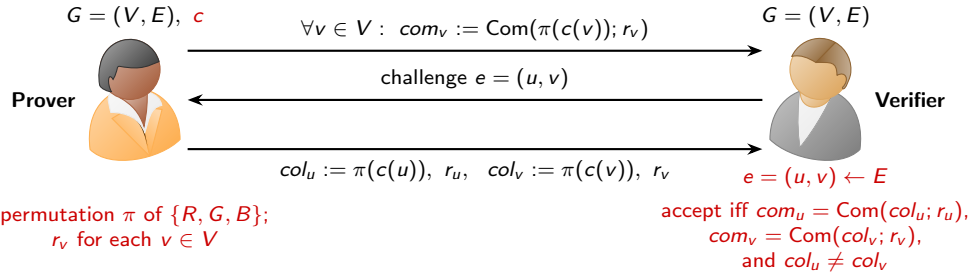
1. $b \leftarrow \{0,1\}$.
2. If $b=0$: view := the view of V^* in $P(1^n,x,w) \leftrightarrow V^*(1^n,x)$ (real)
   If $b=1$: view $\leftarrow \mathcal{S}^{V^*}(1^n,x)$ (simulated)
3. $\mathcal{D}$ is given 1^n and view, and outputs b'. Output 1 if $b'=b$.

Definition 4 (Zero-knowledge)

(P,V) is **zero-knowledge** if there is a PPT simulator $\mathcal{S}$ such that for all PPT V^*, all x,w with $C(x,w)=1$, and all PPT $\mathcal{D}$:

$$\Pr\left[\mathrm{ZK}_{\mathcal{D},V^*,\mathcal{S}}(n,x,w)=1\right] \leq \tfrac{1}{2} + \mathrm{negl}(n).$$

A Zero-Knowledge Proof for Three-Coloring



The colors are **shuffled** by a fresh π every time, and only the two endpoints of one edge are opened.

Three-Coloring: Completeness and Soundness

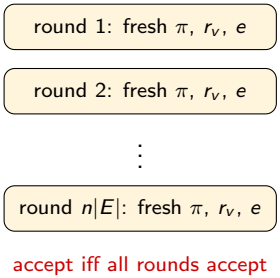
- **Completeness**: $c(u) \neq c(v)$, so $\pi(c(u)) \neq \pi(c(v))$, and the verifier accepts.
- **Soundness**: suppose G is **not** three-colorable, and let col_v be the color committed in com_v (unique, by binding).
  - Since G is not three-colorable, some edge $e = (u, v) \in E$ has $col_u = col_v$.
  - The verifier challenges this edge with probability $1/|E|$, and then rejects.

  So a cheating prover is caught with probability at least $\frac{1}{|E|}$: it succeeds with probability at most $1 - \frac{1}{|E|}$.

Not negligible!

A prover who cheats on one edge gets away with it most of the time. We need to **amplify** soundness.

Soundness Amplification

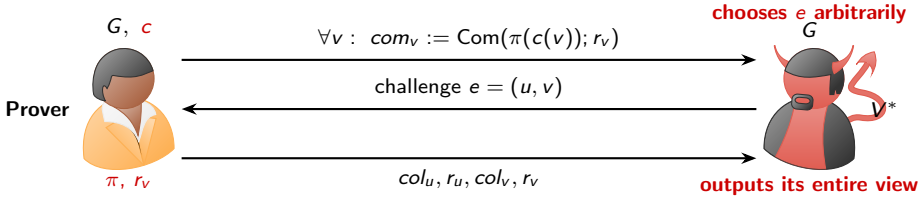


- Repeat the protocol $n|E|$ times, sequentially, with fresh randomness.
- A cheating prover passes each round with probability $\leq 1 - \frac{1}{|E|}$.
- It passes all $n|E|$ rounds with probability

$$\leq \left(1 - \frac{1}{|E|}\right)^{n|E|} \leq e^{-n},$$

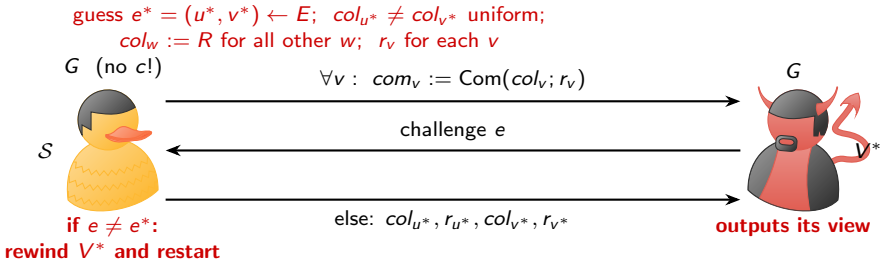
  which is negligible in n (using $1 - z \leq e^{-z}$).
- Completeness is unaffected: an honest prover passes every round.

A Malicious Verifier



- The verifier is now malicious: it may pick e depending on the commitments, and output anything.
- The simulator must produce an indistinguishable output — **without** knowing the coloring c.

The Simulator $\mathcal{S}$



- By hiding, V^* cannot tell which edge $\mathcal{S}$ guessed: $\Pr[e = e^*] \approx 1/|E|$. When this happens, the output of V^* looks like its output with an honest prover.
- So $\mathcal{S}$ runs V^* about $|E|$ times to get an output (stop after $n|E|$ tries: fails with probability $\leq e^{-n}$).

Simulation by Cropping Undesirable Parts



- Great skill?
- Took 156 attempts — the failed ones were cropped.
- Hard to distinguish from the real thing. That is exactly what $\mathcal{S}$ does with its restarts.

The Simulator's Output Is Indistinguishable

	V^* interacts with ...	why it looks the same
H_0	the honest prover (real view)	
H_1	the prover, who also guesses $e^* \leftarrow E$ and **restarts** if $e \neq e^*$	identical: restarting just crops runs (cropping argument)
H_2	... and commits to R for all vertices except u^*, v^*	**hiding** of Com: those commitments are never opened
H_3	the simulator $\mathcal{S}$: $col_{u^*} \neq col_{v^*}$ uniform, no c used	identical: $(\pi(c(u^*)), \pi(c(v^*)))$ is a uniform pair of distinct colors

- $H_0 \approx H_3$: the real view and the simulated view are indistinguishable. **Zero-knowledge**. □
- The step $H_1 \to H_2$ is a reduction to hiding, one commitment at a time (a hybrid over the $|V| - 2$ unopened vertices, as in lecture 6).

Summary

- **Commitments**: **binding** and **hiding**; never both perfect.
- **Constructions**: from a OWP; Pedersen $g^x h^r$.
- **Zero-knowledge**: prove $\exists w : C(x, w) = 1$, reveal nothing else.
- **Three-coloring**: commit to a shuffled coloring, open one edge; repeat.
- **Simulator**: guess the edge, **restart** on a wrong guess.